

Chapter 8 – Google Workspace Directory Management & Admin SDK

Learning Objectives

After completing this chapter, you should be able to:

- Understand the Google Admin SDK.
 - Explain the Directory API.
 - Understand service accounts.
 - Explain Domain-Wide Delegation (DWD).
 - Choose the correct OAuth scopes.
 - Understand Admin SDK architecture.
 - Design secure enterprise automations.
 - Troubleshoot common Admin SDK issues.
-

1. Business Problem

Imagine a company with:

- 40,000 employees
- 150 new hires every week
- 60 resignations every week
- Hundreds of department transfers
- Thousands of group membership changes

Creating users manually in the Admin Console is not practical.

The company wants automation.

For example:

- HR hires an employee.
- The employee account is created automatically.
- The correct license is assigned.
- The employee is placed in the correct OU.
- Required groups are assigned.
- A welcome email is sent.

This is where the **Admin SDK** comes in.

2. What is the Admin SDK?

The Google Admin SDK is a collection of APIs that allow administrators and applications to manage Google Workspace programmatically.

Instead of clicking through the Admin Console,

applications send API requests.

Examples:

- Create users
- Suspend users
- Update users
- Manage groups
- Manage aliases
- Retrieve audit reports

Admin Console vs Admin SDK

Admin Console	Admin SDK
Manual administration	Automated administration
Web interface	REST APIs
Suitable for small changes	Suitable for large-scale automation
Human administrator	Applications and scripts

3. Admin SDK APIs

The Admin SDK includes several APIs.

API	Purpose
Directory API	Manage users, groups, OUs, domains, aliases
Reports API	Retrieve audit and activity reports
Licensing API	Manage Workspace licenses
Role Management (via Directory API resources)	Manage admin roles and assignments

The **Directory API** is the most commonly used.

4. Directory API

The Directory API manages identity objects inside Google Workspace.

It can:

- Create users
- Update users
- Suspend users
- Delete users
- Restore users
- Manage groups
- Manage aliases
- Manage Organizational Units
- Retrieve user information

Think of it as the programmatic equivalent of the **Directory** section in the Admin Console.

Directory Objects

Google Workspace Directory

|

├─ Users

├─ Groups

├─ Organizational Units

├─ Domains

├─ Aliases

└─ Admin Roles

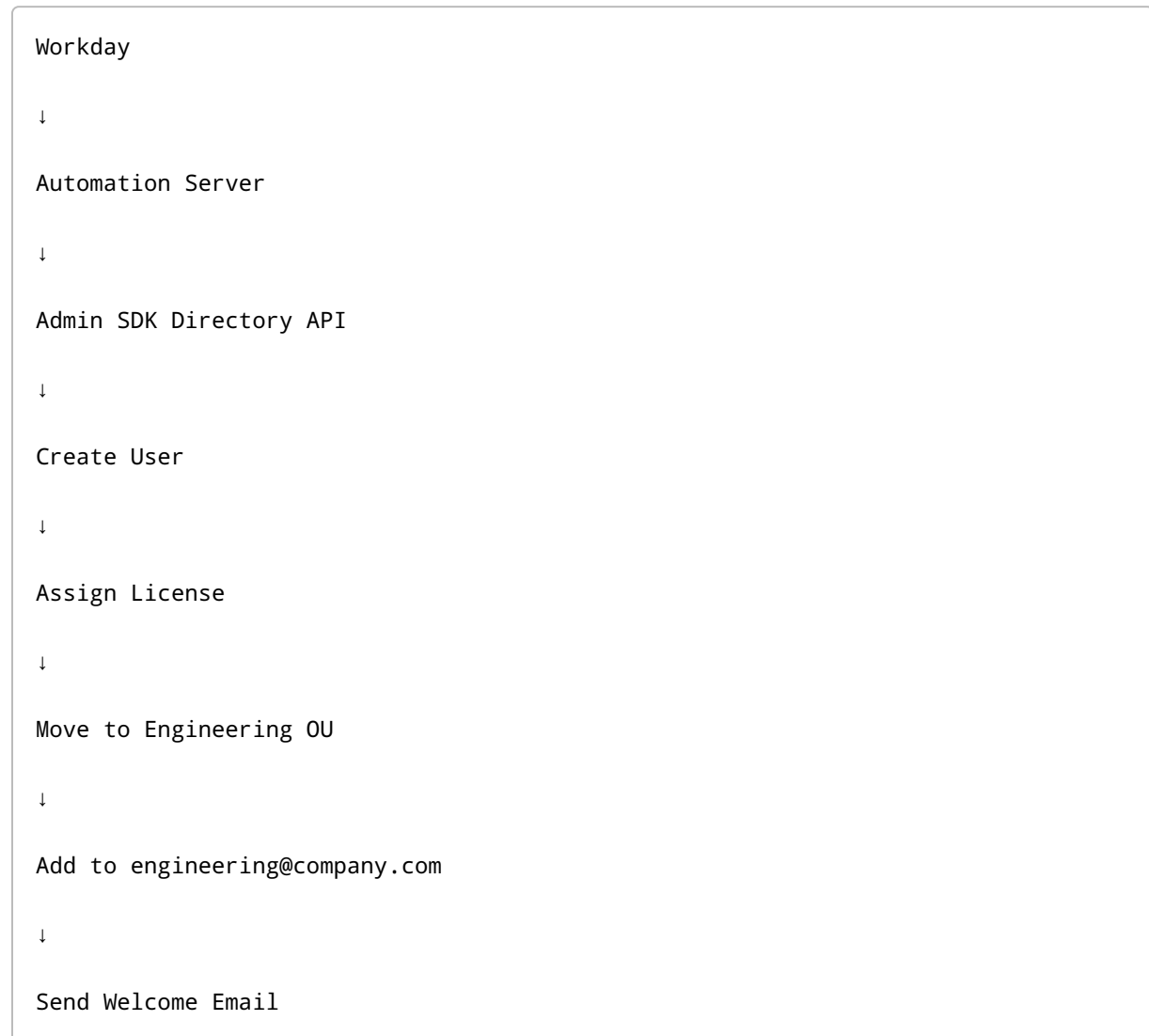
Each object can be managed through API calls.

5. Example Automation

HR hires:

Rahul Sharma

Automation Flow



No administrator manually creates the account.

6. Service Accounts

A service account is a special Google identity used by applications instead of people.

Characteristics:

- Represents an application
- Does not sign in interactively like a user
- Uses credentials to authenticate
- Can perform automated tasks

Examples:

- HR onboarding automation
- User synchronization
- Reporting scripts
- Backup utilities

User Account vs Service Account

User Account	Service Account
Represents a person	Represents an application
Interactive login	Programmatic authentication
Uses password, passkey, or MFA	Uses application credentials or delegated identity
Human activities	Automated processes

7. Domain-Wide Delegation (DWD)

Normally,

an application can only access data after a user grants consent.

Enterprise automation is different.

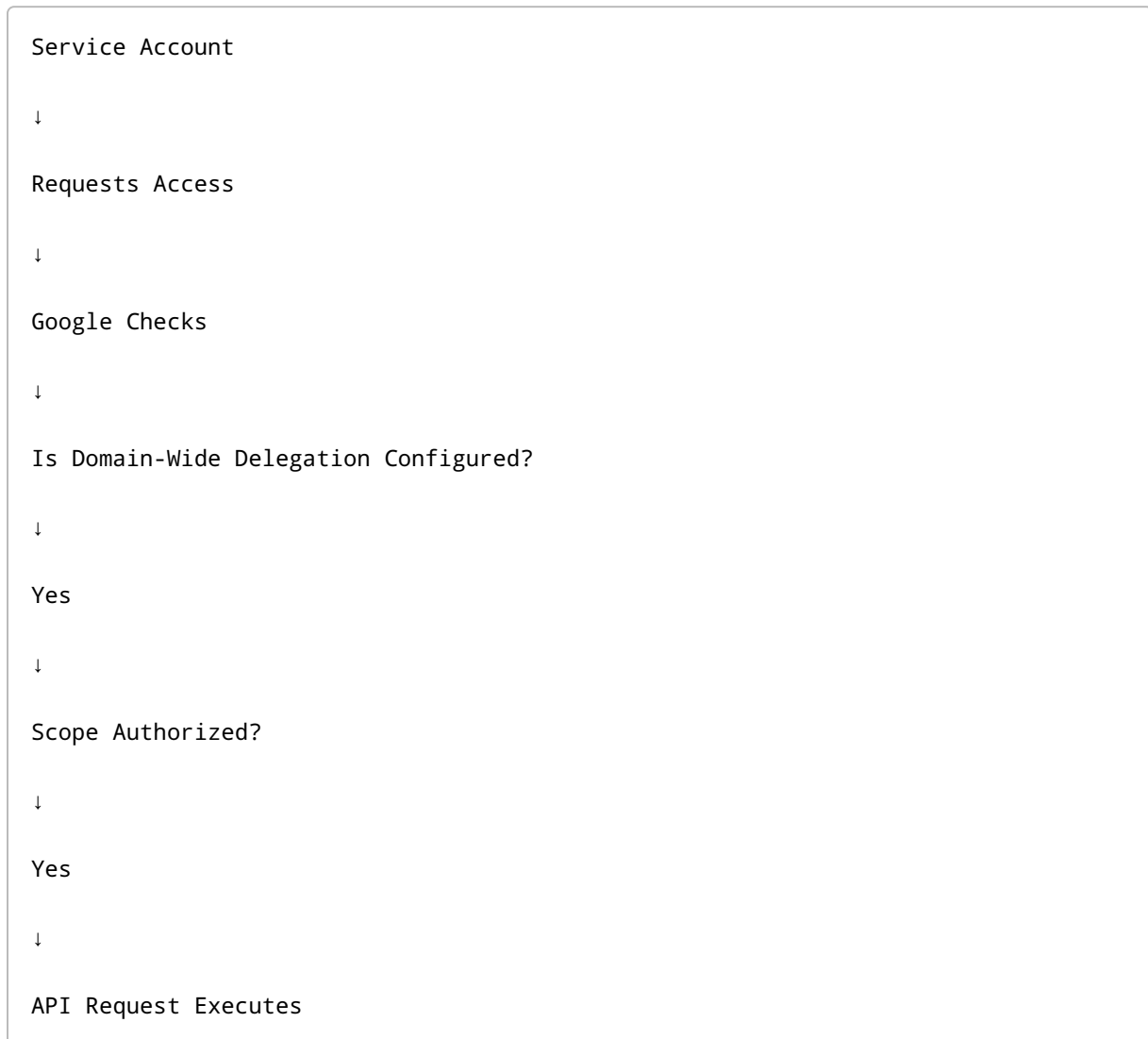
Imagine onboarding 500 employees.

Waiting for each employee to approve consent is impossible.

With Domain-Wide Delegation,

a Workspace administrator authorizes a service account to act on behalf of users for approved scopes.

Domain-Wide Delegation Flow



Without DWD,

many Admin SDK operations cannot be performed automatically.

8. OAuth Scopes

Scopes define what the service account can do.

Examples:

Scope	Purpose
Admin SDK Directory	Manage users and groups
Reports API	Read audit logs
Gmail API	Send or manage email
Drive API	Manage Drive files
Calendar API	Manage calendar events

Grant only the scopes required.

Example

User onboarding automation

Needs:

- Create user
- Update user
- Add user to group

Required Scope

✓ Admin SDK Directory API

Not Required

✗ Gmail API

✗ Drive API

✗ Calendar API

9. Least Privilege Design

Suppose a company has three automation workflows.

Automation 1

Purpose

User onboarding

Service Account

Onboarding-SA

Scopes

- Directory API
-

Automation 2

Purpose

Send welcome emails

Service Account

Mail-SA

Scopes

- Gmail API
-

Automation 3

Purpose

Monthly reporting

Service Account

Reports-SA

Scopes

- Reports API

Each automation has its own service account.

This limits the impact if one account is compromised.

10. Enterprise Architecture

HR System

↓

Automation Platform

↓

Service Account

↓

Domain-Wide Delegation

↓

Google Admin SDK

↓

Google Workspace

This architecture is common in enterprise environments.

11. Common Admin SDK Errors

403 Forbidden

Cause:

The request is authenticated,

but not authorized.

Possible reasons:

- Missing Domain-Wide Delegation
- Required scope not authorized
- Service account lacks permissions
- Admin role is insufficient

Example

A service account calls the Directory API.

Domain-Wide Delegation was not configured.

Result:

403 Forbidden

401 Unauthorized

Cause:

Authentication failed.

Possible reasons:

- Invalid credentials
 - Expired token
 - Incorrect service account configuration
-

Insufficient Permissions

Cause:

The service account authenticated successfully,
but lacks the required OAuth scope or delegated privileges.

Resolution:

Review:

- Authorized scopes
 - Admin role
 - Domain-Wide Delegation configuration
-

User Not Found

Cause:

The requested user does not exist.

Possible reasons:

- Incorrect email address
- User deleted
- Provisioning delay

12. Troubleshooting

When an Admin SDK automation fails,

follow a structured approach.

1. Did the service account authenticate successfully?
2. Is Domain-Wide Delegation enabled?
3. Is the required OAuth scope authorized?
4. Does the service account have the necessary admin privileges?
5. Does the target user exist?
6. Is the API enabled?
7. Are there organization policies restricting the action?

Avoid assuming every error is a coding problem.

Many issues are configuration or authorization related.

Enterprise Scenario

An HR onboarding application creates Google Workspace accounts.

The workflow suddenly fails.

Error:

403 Forbidden

Investigation:

- Service account authenticates successfully.
- Domain-Wide Delegation is enabled.
- Directory API scope was accidentally removed from the authorized scopes.

Resolution:

Restore the required Directory API scope.

Automation resumes successfully.

Best Practices

- ✓ Use one service account per business function.
 - ✓ Authorize only the required OAuth scopes.
 - ✓ Review Domain-Wide Delegation regularly.
 - ✓ Audit service account usage.
 - ✓ Monitor Admin SDK logs and failures.
 - ✓ Rotate credentials or prefer keyless authentication methods where possible.
 - ✓ Document every automation workflow.
-

Common Mistakes

- ✗ Using one service account for every automation.
 - ✗ Granting unnecessary scopes.
 - ✗ Forgetting to configure Domain-Wide Delegation.
 - ✗ Assuming authentication automatically grants authorization.
 - ✗ Ignoring API error messages.
-

Interview Questions

1. What is the Google Admin SDK?
 2. What is the Directory API?
 3. Why would you use the Admin SDK instead of the Admin Console?
 4. What is a service account?
 5. What is Domain-Wide Delegation?
 6. Why is the Principle of Least Privilege important for service accounts?
 7. What causes a 403 Forbidden error when calling the Directory API?
 8. How would you design a secure onboarding automation?
-

Key Takeaways

The Admin SDK enables programmatic management of Google Workspace.

The Directory API manages users, groups, Organizational Units, domains, and aliases.

Service accounts allow applications to automate administrative tasks.

Domain-Wide Delegation enables approved service accounts to act on behalf of users without individual consent.

Always follow the Principle of Least Privilege by granting only the required OAuth scopes and separating automations by function.

Chapter Summary

In this chapter you learned:

- ✓ What the Admin SDK is.
- ✓ The purpose of the Directory API.
- ✓ Service accounts.

✓ Domain-Wide Delegation.

✓ OAuth scopes.

✓ Enterprise automation architecture.

✓ Common Admin SDK errors and troubleshooting.

The next chapter will cover **Gmail Architecture and Mail Flow**, where you'll learn how email travels across the internet, including DNS, MX records, SPF, DKIM, DMARC, SMTP, routing, and common mail delivery troubleshooting—connecting directly to the scenarios we've already discussed.